Geometry Name____________________
Unit 9 Date_____________________
Lesson 5 Homework Period____________________

Using the given three figures, answer questions 1-3.

1. Which two figures can have the same volume?

2. Explain how the figures listed in Question 1 can have the same volume.

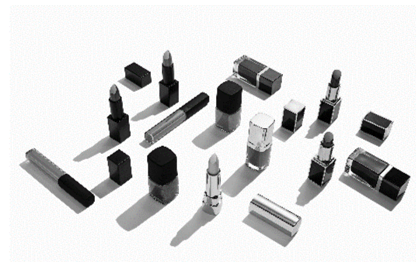
3. What information is required for each figure to determine if the volumes are equal?

4. Isaac and Dallas went to an amusement park. They each bought a cheeseburger. Isaac's meat patty came with a square base. Dallas's meat patty had a circular base. Is it possible that they both purchased the same size of cheeseburger?

5. Dallas's meat patty was 0.75 inches high with a radius of 4.5 inches. What is the volume?

6. Isaac's popcorn container was also 11.5 inches high but with a base whose side length was 4.5 inches. Determine if the two containers have the same volume.

7. Brand A nail polish is sold in rectangular prism jars while Brand B nail polish is sold in cylinder jars. Is it possible for a given quantity of Brand A nail polish and Brand B nail polish to have the same volume?



8. A jar of Brand B nail polish is 3 inches tall, with a diameter of 1.25 inches. In a store display, there are a total 18 jars. What is the volume of nail polish in the display?

9. One jar of Brand A nail polish is 3 inches tall, with a width of 1 inch and a length of 1.25 inches. There are 18 jars in the display. What is the total volume of nail polish in the display?

10. Compare the answers from Questions 8 and 9.